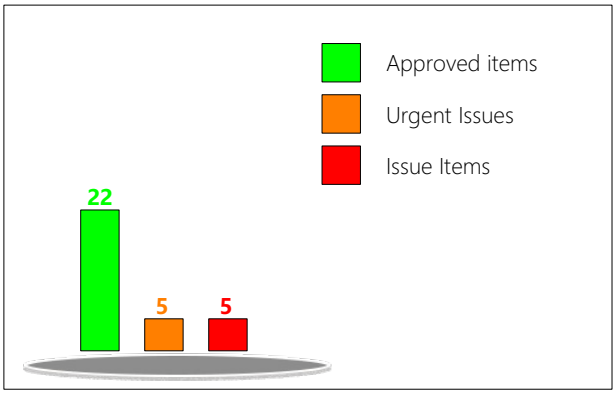
HQDFM Design for Manufacture(DFM) Report

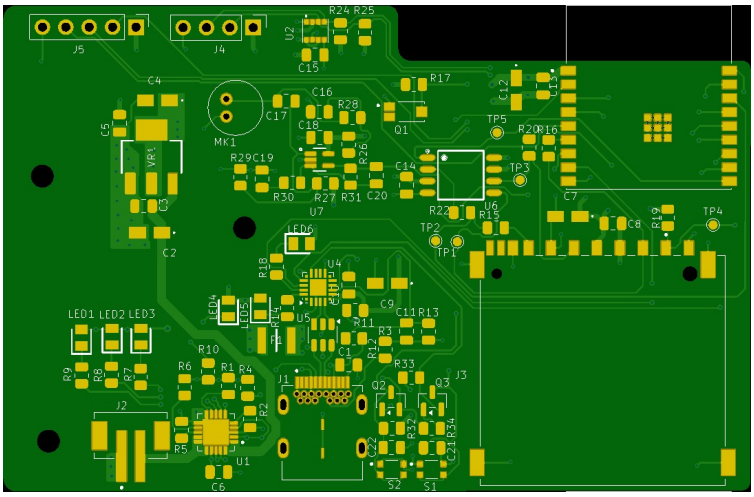
File name: gerber

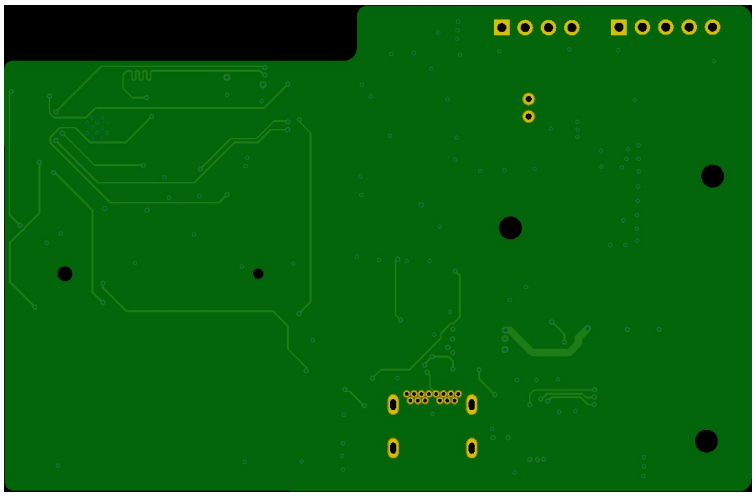
Time: 2026-10-09 Layer count:5 PCB Thickness: 1.60 Quantity: 5 mm



Basic Board Specs	Trace Width/Spacing	8.00/8.00mil
	Milling Density	63.2041m/m²
	Surface Finish Area	11.35%
	Test Point Count	311

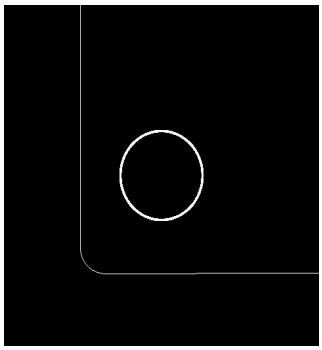
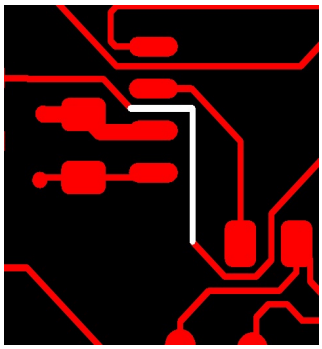
The file size is small, which can affect the surface mount assembly process. It is recommended to have a size larger than 7*7cm. You can optimize the size by adding a process edge or increasing the panelization.

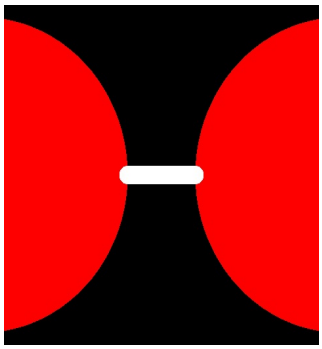
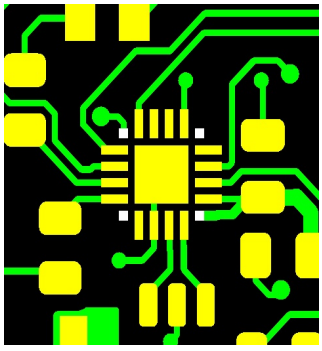


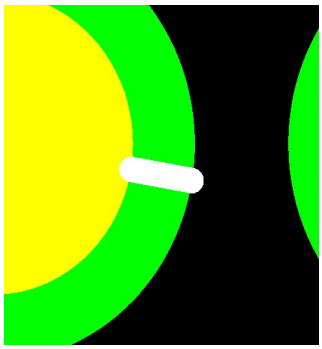
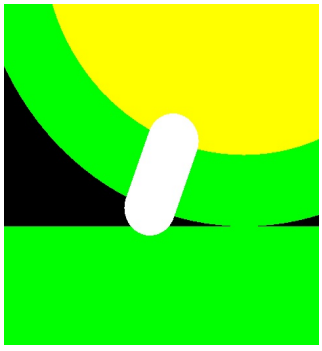


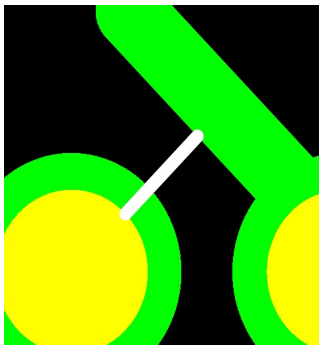
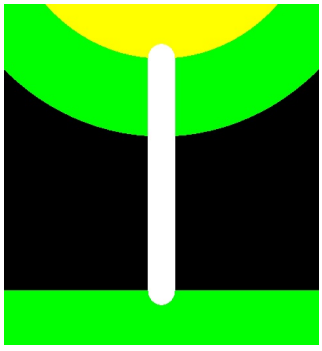
Type	Category	No. of Checks	Result
PCB Trace Analysis	Open/Shorts (IPC)	1	Fail
	Signal Integrity	4	Fail 4
	Smallest Trace Width	1	Pass 3
	Smallest Trace Spacing	3	Pass 683 , Fail 36
	SMD Pad Spacing	1	Pass
	Pad Size	3	Pass 30 , Fail 1
	Hatched Copper Pour	2	Pass
	Annular Ring Size	2	Pass 4 , Fail 4
	Drill to Copper	5	Pass 165 , Fail 14
	Copper-to-Board Edge	2	Pass 41
	Holes on SMD Pads	4	Pass
PCB Drilling Analysis	Drill Diameter	8	Pass 18 , Fail 1
	Drill Spacing	4	Pass 34 , Fail 11
	Drill to Board Edge	4	Pass 2
	Drill Hole Density	1	Pass
	Special Drill Holes	2	Pass
	Drill Hole Errors	3	Pass
PCB Solder Mask Analysis	Solder Mask Dam	3	Pass 29 , Fail 2
	Missing SMask Opening	1	Pass
	Solder Paste Area	1	Pass
PCB Silk Analysis	Silkscreen Spacing	1	Pass 5 , Fail 5

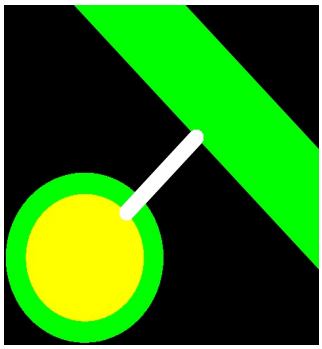
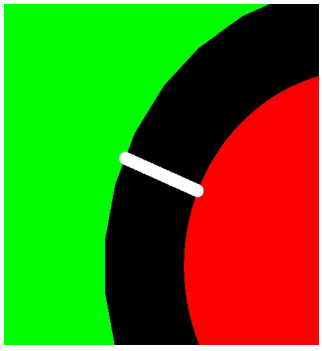
PCBA Component Analysis	Component Spacing	2	Fail
	Comp Height Analysis	3	Pass
	Comp.-to-Board-Edge	3	Fail
	Component Silkscreen Spacing	0	Fail
	Pad Count Mismatch	4	Fail
	Designator Length	0	Fail
	Double-sided Components	1	Pass
	Mid Mount Comp	1	Pass
PCBA Pin Analysis	SMT Pad Analysis	8	Fail
	Through-hole Pins	10	Fail
PCBA Pad Analysis	Chip Pads	60	Fail
PCBA Fiducial Analysis	Fiducial Count	1	Fail
	Fiducial Analysis	4	Pass

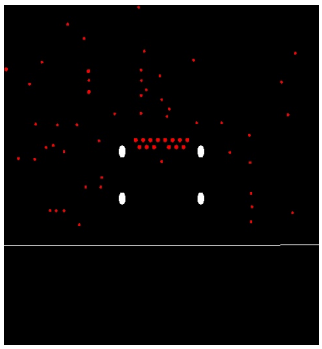
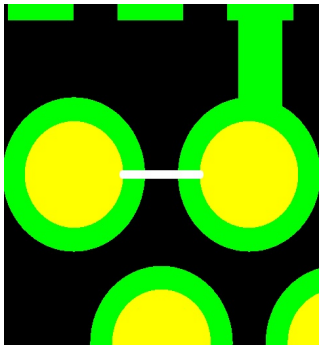
ID	Check	Limits	Value	Issue	Image	Position	Qty	Level
1	Signal Integrity_Floating Copper	-,-,-	Error(s) detected	Isolated areas of copper (floating copper) were detected in your design. Leaving isolated areas of copper unconnected can cause them to act as antennas and can affect signal integrity. We suggest removing or grounding them.		99.85,-107.00	3	Risk
2	Signal Integrity_Acute Angle Traces	-,-,-	Error(s) detected	There is an acute angle connection mode in your design,which will affect the signal integrity of the product.it is recommended to adjust the acute angle posintion to arc or obtuse angle connection mode		141.32,-80.10	1	Risk

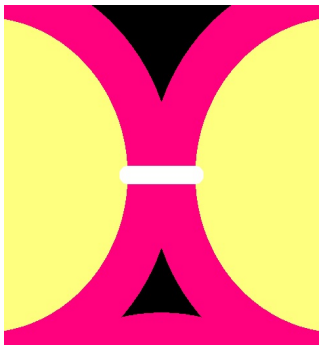
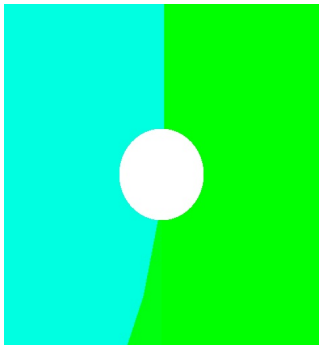
3	Smallest Trace Spacing_SMD Pad Spacing	4,6,10	0.15 mm	Pad-to-pad spacing of 5.91mil were detected in your design. This could result in incomplete solder mask dams between the pads, thereby increasing the risk of incomplete etching of the pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 6 mill. For solder mask covered pads, 4 mil spacing is sufficient.		127.63,-102.59	28	Warning
4	Pad Size_Short Pads	8,12,20	0.30 mm	Regular pads of size 0.30mm were detected in your design. This could result in overetching, detached pads and increased testing costs, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The pads should be enlarged to at least 0.3 mm if possible.		129.15,-90.32	14	Warning

5	Annular Ring Size_PTH Annular Ring	6,7,8	0.05 mm	PTH Pad Rings3.91 mil in size were detected in your design. This could increase the risk of tangency or breakouts, which decrease electrical reliability and yield, and affect the reliability of the boards. It is recommended that PTH Pad Rings have a width of at least 6 mil.		127.10,-101.93	2	Risk
6	Annular Ring Size_Via Annular Ring	4,5,6	0.05 mm	Min via annular rings1.93mil in size were detected in your design.It will affect production efficiency and electrical reliability. It is recommended that the minimum ring size for "via annular rings" be ≥5 mils.		164.90,-73.69	6	Risk

7	Drill to Copper_PTH-to-Trace [Inner]	10,12,14	0.25 mm	PTH to trace spacing (inner layer) of 11.79mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		129.50,-101.61	19	Warning
8	Drill to Copper_PTH-to-Trace [Outer]	10,12,14	0.25 mm	PTH to trace spacing (outer layer) of 11.87mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		131.23,-102.96	22	Warning

9	Drill to Copper_Via-to-Trace [Outer]	8,10,12	0.25 mm	Via-to-trace spacing (outer layer) of 9.83mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 10 mil.		166.81,-71.68	105	Warning
10	Drill to Copper_NPTH-to-Copper	8,10,12	0.25 mm	NPTH to copper spacing of 9.81mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		147.93,-88.55	8	Warning

11	Drill Diameter_Slot Aspect Ratio	1.5,2,2	0.30 mm	Slots with aspect ratio of 1.92 were detected in your design. This could increase the risk of incomplete drilling of the slot, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The ratio should be increased to at least 2:1		129.63,-105.39	1	Warning
12	Drill Spacing_Different Net PTH Spacing	0.4,0.45,0.5	0.35 mm	For most factories, the different net PTH spacing requirement is 16-18 mil. Failure to meet the factory's requirements could increase the risk of solder bridges during soldering, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Different net PTH spacing of 13.70mil were detected in your design. It is recommended to increase the spacing to at least 17 mil.		130.43,-101.88	11	Risk

13	Solder Mask Dam_Solder Mask Dam	5,6,10	5.91 mil	Solder mask dam of width less than 5.91mil were detected in your design. This could result in incomplete solder mask dams between the pads, thereby increasing the risk of solder bridges during assembly, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The width should be at least 6 mil where possible.		128.03,-101.88	2	Warning
14	Silkscreen Spacing_Solder Mask-to-Silkscreen	4,5,6	Error(s) detected	For most factories, the minimum silkscreen to solder mask spacing requirement is at least 8 mil. Failure to meet the factory's requirements could result in part of the silkscreen being removed or being printed directly on the pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Silkscreen to solder mask spacing of 0 mil were detected in your design. It is recommended to increase the spacing to at least 12 mil.		154.33,-67.76	10	Risk